Power BI Practical Tutorial

NovaMart Retail Sales Dashboard - Guided Student Activity

Objective: Use the supplied NovaMart Excel dataset to clean data, create calculations, build a simple interactive dashboard, and answer business questions.

Dataset	NovaMart_PowerBI_CaseStudy_Dataset.xlsx
Suggested time	90-120 minutes
Output	One-page Power BI dashboard + short analysis
Skills	Power Query, DAX, Visuals, Slicers, Business Insights

Step 1 - Import the Dataset

1. Open **Power BI Desktop**.
2. Choose **Home > Get Data > Excel Workbook**.
3. Select the supplied NovaMart Excel file.
4. Select **Sales_Data**.
5. Click **Transform Data**.

Step 2 - Inspect the Data

In Power Query enable **View > Column Quality, Column Distribution, and Column Profile**. Look for duplicate Order IDs, missing Salesperson values, inconsistent Region/Product Category values, and incorrect data types.

Step 3 - Check Data Types

Order Date = Date; Quantity = Whole Number; Unit Price and Unit Cost = numeric; descriptive columns = Text.

Step 4 - Clean the Dataset

Remove duplicate **Order ID** records. Replace null Salesperson values with **Unknown**. Standardize Region labels such as western/CENTRAL/South/North/eastern. Correct **Electronic** to **Electronics** and **Home Appliance** to **Home Appliances**.

Step 5 - Create Calculated Columns

Use **Add Column > Custom Column** to create the following:

Sales Amount

```
[Quantity] * [Unit Price]
```

Total Cost

```
[Quantity] * [Unit Cost]
```

Profit

```
[Sales Amount] - [Total Cost]
```

Select **Home > Close & Apply**.

Step 6 - Create DAX Measures

Create these with **Modeling > New measure**. Do not enter them in DAX Query View.

Total Sales

```
Total Sales =
SUM('Sales_Data'[Sales Amount])
```

Total Cost

```
Total Cost =
SUM('Sales_Data'[Total Cost])
```

Total Profit

```
Total Profit =
SUM('Sales_Data'[Profit])
```

Total Orders

```
Total Orders =
DISTINCTCOUNT('Sales_Data'[Order ID])
```

Total Quantity Sold

```
Total Quantity Sold =
SUM('Sales_Data'[Quantity])
```

Profit Margin %

```
Profit Margin % =
DIVIDE([Total Profit], [Total Sales], 0)
```

Format **Profit Margin %** as Percentage.

Step 7 - Build the Dashboard

Create a page named **Executive Sales Dashboard** and add:

Visual	Field / Measure
Card	Total Sales
Card	Total Profit
Card	Total Orders
Card	Profit Margin %
Line Chart	Order Date (Month) vs Total Sales
Bar Chart	Region vs Total Sales
Column Chart	Product Category vs Total Sales
Bar Chart	Product Category vs Profit Margin %
Bar Chart	Product vs Total Sales; Top N = 5
Column/Donut	Sales Channel vs Total Sales

Step 8 - Add Slicers

Add slicers for **Order Date, Region, Product Category, and Sales Channel**. Test that the visuals update when filters change.

Step 9 - Answer the Business Questions

1. Which region has the highest Total Sales?
2. Does the highest-sales region also have the highest Profit Margin %?
3. Which Product Category generates the highest revenue?
4. Which category has high sales but weaker profitability?
5. Is Online or Physical Store more profitable?
6. Which month has the strongest performance?
7. Which three products should management prioritize?
8. Identify one business problem that could be hidden by looking only at Total Sales.

Step 10 - Management Recommendations

Provide **three recommendations**. For each one use:

Finding - what you discovered.
Evidence - the KPI/chart that supports it.
Action - what management should do.

Submission Checklist

- Cleaned Power Query dataset
- Three calculated columns
- Six DAX measures
- Four KPI cards
- Required charts
- Four working slicers
- Answers to eight business questions
- Three management recommendations
- Screenshot of final dashboard

Learning flow: Raw Data -> Cleaning -> Calculations -> Dashboard -> Insight -> Business Decision.